

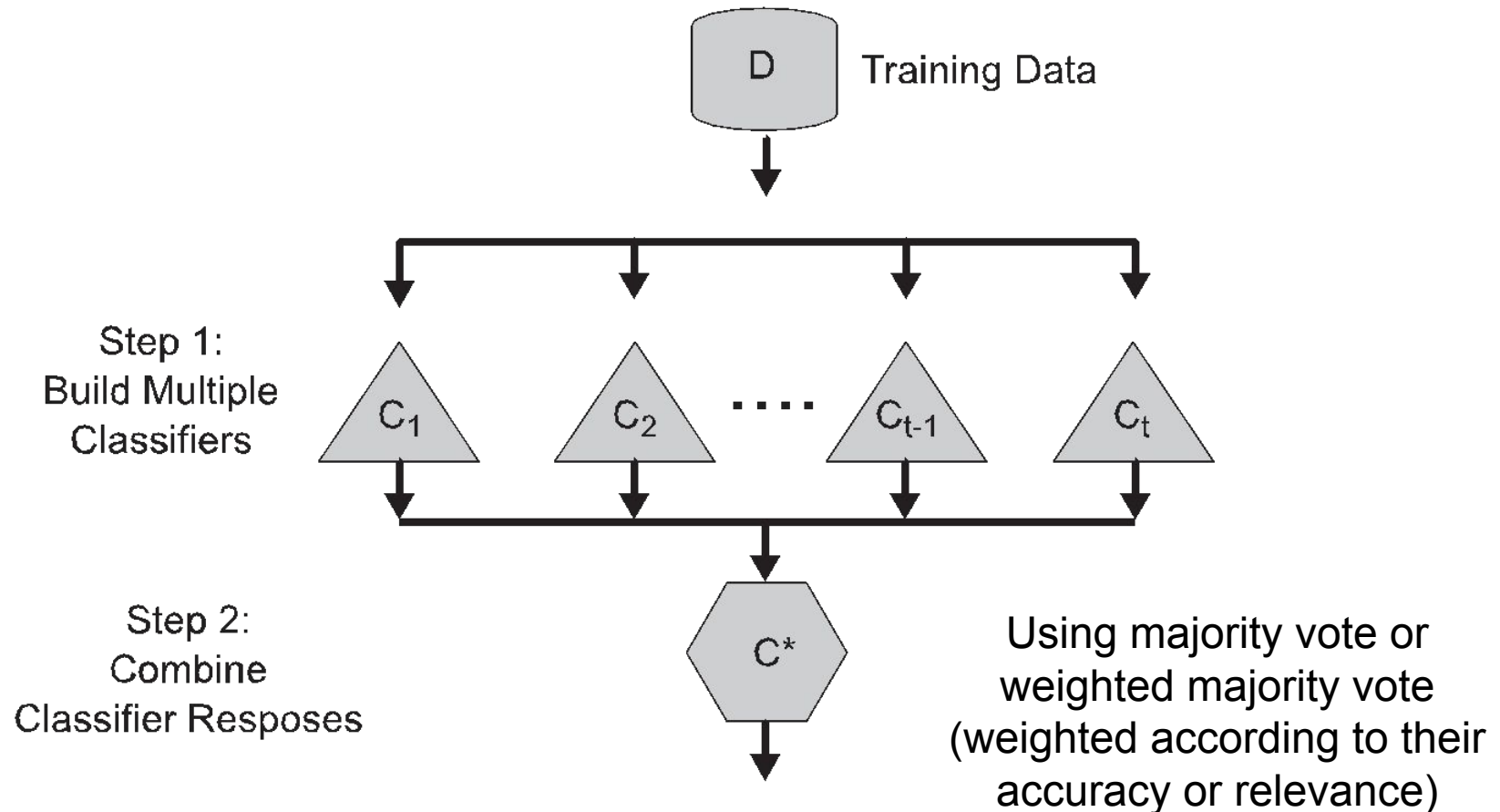


Ensemble Techniques

Ensemble Methods

- Construct a set of base classifiers learned from the training data
- Predict class label of test records by combining the predictions made by multiple classifiers (e.g., by taking majority vote)

General Approach of Ensemble Learning



Constructing Ensemble Classifiers

- By manipulating training set
 - Example: bagging, boosting, random forests
- By manipulating input features
 - Example: random forests
- By manipulating class labels
 - Example: error-correcting output coding
- By manipulating learning algorithm
 - Example: injecting randomness in the initial weights of ANN

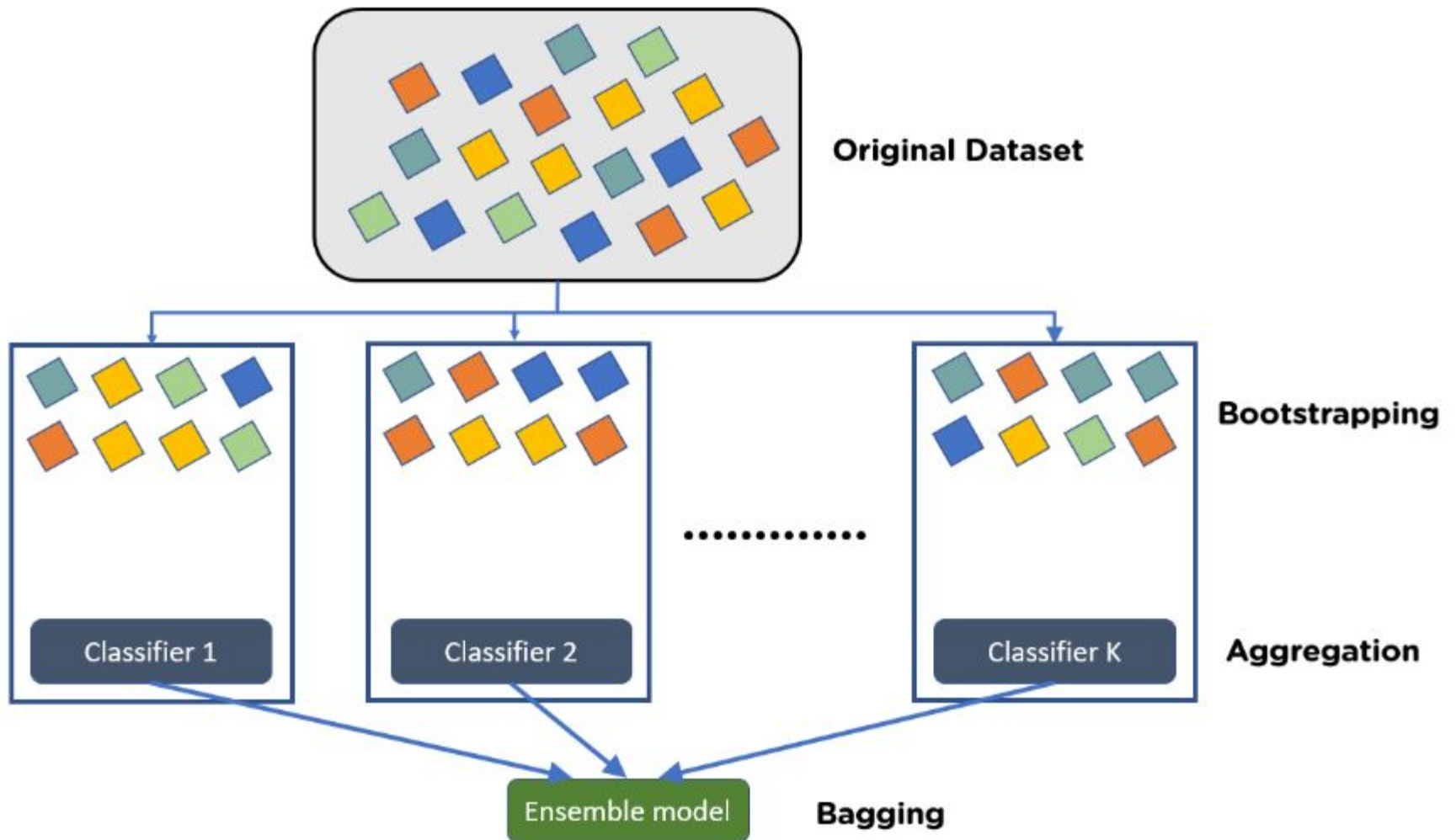
Bagging (Bootstrap AGGregatING)

- Bootstrap sampling: sampling with replacement

Original Data	1	2	3	4	5	6	7	8	9	10
Bagging (Round 1)	7	8	10	8	2	5	10	10	5	9
Bagging (Round 2)	1	4	9	1	2	3	2	7	3	2
Bagging (Round 3)	1	8	5	10	5	5	9	6	3	7

- Build classifier on each bootstrap sample
- Probability of a training instance being selected in a bootstrap sample is:
 - $1 - (1 - 1/n)^n$ (n: number of training instances)
 - ~ 0.632 when n is large

How Bagging Works



Boosting

- An iterative procedure to adaptively change distribution of training data by focusing more on previously misclassified records
 - Initially, all N records are assigned equal weights (for being selected for training)
 - Unlike bagging, weights may change at the end of each boosting round

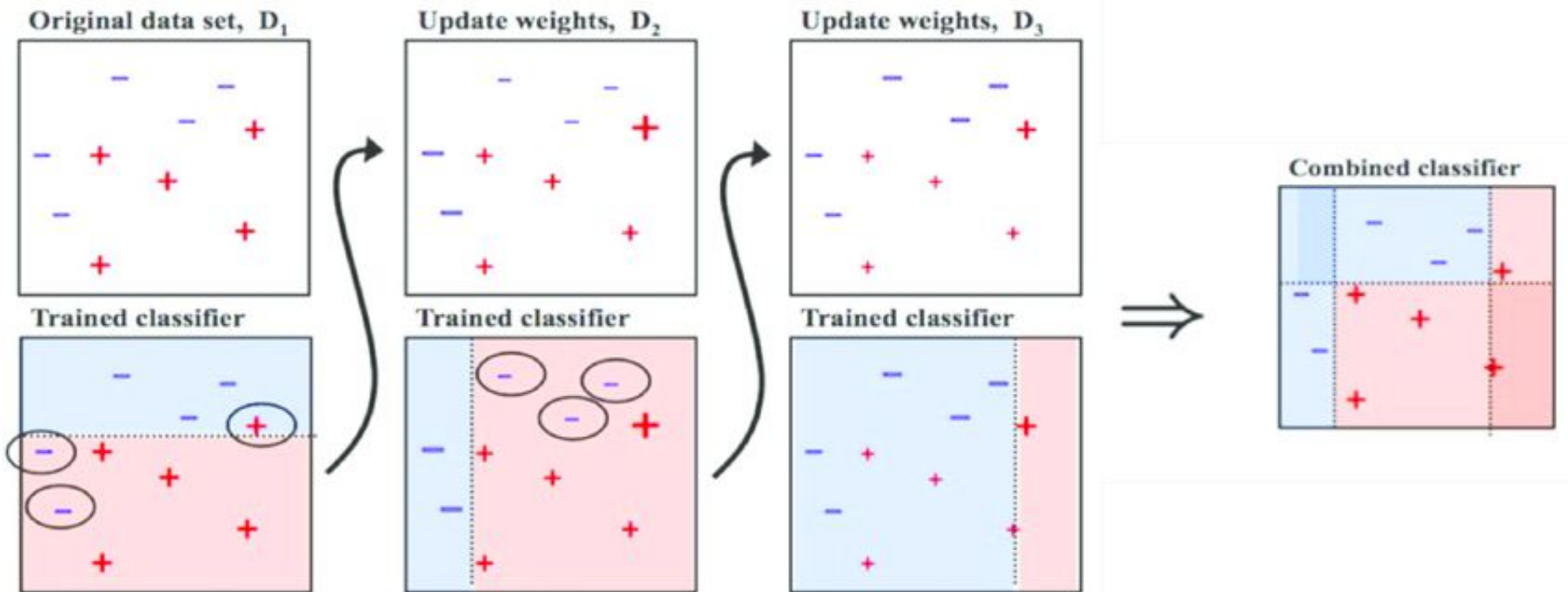
Boosting

- Records that are wrongly classified will have their weights increased in the next round
- Records that are classified correctly will have their weights decreased in the next round

Original Data	1	2	3	4	5	6	7	8	9	10
Boosting (Round 1)	7	3	2	8	7	9	4	10	6	3
Boosting (Round 2)	5	4	9	4	2	5	1	7	4	2
Boosting (Round 3)	4	4	8	10	4	5	4	6	3	4

- Example 4 is hard to classify
- Its weight is increased, therefore it is more likely to be chosen again in subsequent rounds

How Boosting Works



Random Forest Algorithm

- Construct an ensemble of decision trees by manipulating training set as well as features
 - Use bootstrap sample to train every decision tree (similar to Bagging)
 - Use the following tree induction algorithm:
 - ◆ At every internal node of decision tree, randomly sample p attributes for selecting split criterion
 - ◆ Repeat this procedure until all leaves are pure (unpruned tree)

How Random Forest Works

